

ROUTE 1 (PATH 1) — COMPLETE ORGANIZER ANSWER KEY & OPERATIONS GUIDE

ROUTE: ECE → MECH → FOOD COURT → CYBER SECURITY → AUDI

⚠ ORGANIZER INSTRUCTION: This official manual corresponds to PATH1_FINAL_ODK.xlsx. All typed answers strictly enforce **MANDATORY UPPERCASE**.

1. MASTER FLOW & VOLUNTEER PASS CODES

Stage	Challenge Description	Required Answer / Code	Issued By	Unlocks Next
1. ECE Block	Logic Gate Quiz & Hardware Hunt	P1-ECE-PASS	ECE Volunteer (Top 25)	NEXT BLOCK: MECH
2. MECH Sound	Listen to engine_sound.wav → ENGINE	P1-MECH-START	MECH Volunteer	MECH Mini-Tasks
3. MECH Mini-Tasks	Human Shape • Garland • Bomb Defusal	P1-MECH-PASS	MECH Volunteer (Top 15)	NEXT BLOCK: FOOD COURT
4. Food Court	Number Shuffle (56 Qs across Sets A–G)	P1-FC-PASS	FC Volunteer (Top 7)	NEXT BLOCK: CY
5. Cyber Security	5 Caesar Phases (-5 Shift) & Agility	P1-CY-PASS	CY Volunteer (Top 5)	NEXT BLOCK: AUDI
6. AUDI Stage	Auditorium Riddle → Stage → Final Physical	FINAL-PATH1	Chief Judges	🏆 PATH 1 COMPLETE

2. ECE CHALLENGES & HARDWARE HUNT BANK

Variant	Technical Question & Solution	Building Hunt Discovered Keyword
Variant A	Which passive component opposes sudden changes in electric current? → INDUCTOR	INTEGRATION
Variant B	What is the base frequency unit named after a German physicist (abbreviation Hz)? → HERTZ	MODULATION
Variant C	What fundamental semiconductor device allows electric current to flow in only one direction? → DIODE	FREQUENCY
Variant D	In Boolean algebra: what logic gate produces a HIGH output only when both inputs are HIGH? → AND GATE	OSCILLATOR
Variant E	What property of a material characterizes its opposition to the flow of electric charge? → RESISTANCE	WAVELENGTH
Variant F	Which basic electronic component stores electrical energy in an electrostatic field? → CAPACITOR	AMPLITUDE
Variant G	In communication theory: what is the spread between upper and lower frequencies called? → BANDWIDTH	RESONANCE

3. NUMBER SHUFFLE ANSWER BANK (56 QUESTIONS ACROSS 7 SETS)

Rule: 4=A, 5=B, 6=C... 29=Z. All 8 questions per set must be solved sequentially in UPPERCASE.

Set	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Set A	HUNGRY	DESSERT	BREAKFAST	CAFETERIA	BEVERAGE	APPETITE	CANTEEN	FOOD COURT
Set B	SANDWICH	NOODLES	PIZZA	BURGER	SALAD	COFFEE	TEA	SNACKS
Set C	JUICE	WATER	BISCUIT	CHOCOLATE	ICE CREAM	PASTA	MANGO	SPICE
Set D	GRILL	TOAST	BUTTER	CHEESE	PEPPER	GARLIC	ONION	TOMATO
Set E	LEMON	GINGER	SUGAR	JAM	HONEY	BREAD	RICE	SOUP
Set F	SWEET	SAUCE	DINNER	LUNCH	PLATTER	SERVING	APPETIZER	CUISINE
Set G	DELICACY	NOURISH	TASTY	MEALTIME	CAFETERIA	BEVERAGE	APPETITE	BREAKFAST

4. CAESAR CIPHER DECRYPTION KEY (5 MANDATORY PHASES)

Phase	Ciphertext	Shift Rule	Decoded Answer	Pass Code
Phase 1	FZIN	-5 Backward	AUDI	Phase 2
Phase 2	XYFLJ	-5 Backward	STAGE	Phase 3
Phase 3	WJI XJFYX	-5 Backward	RED SEATS	Phase 4
Phase 4	RNHWMSTSJ	-5 Backward	MICROPHONE	Phase 5
Phase 5	TW NJSYFYNTS	-5 Backward	ORIENTATION	CAESAR-PASS

5. DETAILED CHALLENGE DESCRIPTIONS, MATERIALS & PARALLEL STATION CAPACITIES

Stage 1: ECE Logic & Hardware Circuit Gauntlet

PARALLEL STATIONS: 4 LAB TEST BENCHES

Event & Mechanics: Teams begin at ECE. Solve Boolean logic and circuit questions, then navigate ECE floors to retrieve technical keywords.

Location: ECE Ground Floor Lab	Materials: 4 Test benches, circuit reference sheets	Staffing: 4 Lab Volunteers
Capacity: 25 Starting Teams (Top 25 qualify)	Pass Code: P1-ECE-PASS	Next Block: MECH Block

Stage 2: MECH Sound Clue & Mini-Challenges

PARALLEL STATIONS: 6 WORKSHOP BAYS

Event & Mechanics: Listen to engine_sound.wav in-app, identify ENGINE → MECH. Complete Human Shape (7 poses), Hidden Garland (5 items), Bomb Defusal.

Location: MECH Workshop Bay	Materials: 2 Pose mats, 2 garland rope sets, 2 knot boxes	Staffing: 6 Bay Judges + 1 Head Judge
Capacity: 25 Teams (Top 15 qualify)	Pass Code: P1-MECH-PASS	Next Block: Food Court

Stage 3: Food Court Number Shuffle Challenge

PARALLEL STATIONS: 4 VERIFICATION DESKS

Event & Mechanics: 15 teams decode 8 encrypted food & dining words using 4=A...29=Z rule on mobile devices.

Location: Food Court Foyer	Materials: Number conversion reference sheets	Staffing: 4 Volunteers
Capacity: 15 Teams (Top 7 qualify)	Pass Code: P1-FC-PASS	Next Block: Cyber Security Block

Stage 4: Cyber Security Caesar Cipher & Seminar Agility

PARALLEL STATIONS: 3 DESKS + 2 AGILITY LANES

Event & Mechanics: 7 teams decrypt 5 sequential ciphers (-5 backward) and complete blindfolded agility trial in Seminar Hall.

Location: Cyber Security & Seminar Hall	Materials: Cones, blindfolds, cipher wheels	Staffing: 4 Volunteers
Capacity: 7 Teams (Top 5 qualify)	Pass Code: P1-CY-PASS	Next Block: Main Auditorium

Stage 5: Main Auditorium Grand Finale

PARALLEL STATIONS: 2 STAGE ARENAS

Event & Mechanics: Finalists solve Auditorium riddle (AUDITORIUM), Stage riddle (STAGE), and complete physical coordination trial.

Location: Main Auditorium Stage	Materials: Coordination challenge props	Staffing: 3 Chief Judges
Final Codes: Ans 1: STAGE • Ans 2: FINAL-PATH1	Conclusion: 🏆 PATH 1 COMPLETE	Podium: 1st, 2nd, 3rd Place Award Ceremony